Geospatial Determinants of NCAA Basketball Outcomes: A Replication and Critique of Clay et al. (2015)

CBB Prediction Model Project

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Abstract

We evaluate the claims of Clay, Bro, & Clay (2015) regarding the effects of travel distance, timezone crossing, and other geospatial variables on NCAA basketball outcomes. Using a dataset of 16,689 Division I games (2024–2026) with rich team quality controls, we replicate certain findings—travel distance is a meaningful predictor—but fail to replicate the reported east/west timezone asymmetry. We identify several methodological concerns, most critically the inadequate control for team quality (tournament seed only), double-counting of observations, and post-hoc threshold selection. Our XGBoost model ranks travel distance as feature #8 of 46 by SHAP importance (2.7% of total), while timezone advantage ranks #36 (1.0%). Team quality features account for over 50% of predictive power.

1 Clay et al. (2015): Summary of Methods and Claims

Clay et al. study $N = 1{,}648$ NCAA Tournament games (3,296 team-game observations) from 1985/86 to 2010/11. They estimate a logistic regression:

$$\log\left(\frac{P(\text{win})}{1-P(\text{win})}\right) = \beta_0+\beta_1\cdot\text{TZ}_{\text{east}}+\beta_2\cdot\text{TZ}_{\text{west}}+\beta_3\cdot\Delta\text{elev}+\beta_4\cdot\Delta\text{temp}+\beta_5\cdot D_{150}+\beta_6\cdot\text{seed}+\beta_7\cdot\text{round}+\beta_8\cdot\text{pod} \tag{1}$$

where $\text{TZ}_{\text{east/west}}$ count timezone crossings by direction, D_{150} is a binary indicator for traveling > 150 miles, and seed is the team's tournament ranking (1–16).

Key reported results:

- Traveling east: $\hat{\beta}_1 = -0.150$, OR $= 0.861$, $p = 0.019$
- Traveling west: $\hat{\beta}_2 = -0.065$, OR $= 0.937$, $p = 0.257$
- Distance > 150 mi: $\hat{\beta}_5 = -0.440$, OR $= 0.644$, $p = 0.004$
- Elevation, temperature, pod era: not significant ($p > 0.25$)

2 Methodological Concerns

2.1 Inadequate Control for Team Quality

Clay controls for team strength using tournament seed alone—a 16-level ordinal variable. This is insufficient for two reasons.

First, seed is coarse. A 4-seed and an 8-seed can differ enormously in quality (e.g., a mid-major conference champion seeded 8th versus a power-conference at-large 4-seed). Continuous metrics like Barttorvik's $\hat{\theta}_{\text{barthag}}$ or KenPom efficiency provide far more granular quality control.

Second, seed is confounded with geography. Of the 362 D1 programs in our database:

Timezone	Teams	Share
Eastern (UTC−5)	187	52%
Central (UTC−6)	112	31%
Mountain (UTC−7)	24	7%
Pacific (UTC−8)	38	10%

Table 1: Distribution of D1 programs by timezone.

The Eastern timezone contains a majority of D1 programs and a disproportionate share of power-conference teams. Teams crossing timezones to travel east are disproportionately Mountain/Pacific teams playing against Eastern/Central opponents. Any residual quality differential not captured by the coarse seed variable will be attributed to the timezone crossing.

We tested this directly in our data. For non-neutral games where the away team crosses timezones:

Direction	Away $\bar{\theta}_{\text{barthag}}$	Home $\bar{\theta}_{\text{barthag}}$	Δ(H−A)	N
Same TZ	0.450	0.491	+0.041	10,279
Away → East	0.520	0.566	+0.046	2,038
Away → West	0.506	0.568	+0.062	2,404

Table 2: Team quality by travel direction. Teams traveling west face a larger quality gap.

2.2 Non-Independence of Observations

Clay uses 3,296 "team performances" from 1,648 games. Each game produces two perfectly anti-correlated observations ($Y_{\text{team}_1} = 1 - Y_{\text{team}_2}$). Standard logistic regression assumes independent observations. The effective sample size is $N_{\text{eff}} = 1{,}648$, not 3,296, and standard errors are underestimated by approximately $\sqrt{2}$.

For the timezone-east coefficient ($\hat{\beta}_1 = -0.150$, SE $= 0.064$), correcting for non-independence yields $\text{SE}_{\text{corrected}} \approx 0.064 \times \sqrt{2} = 0.091$, giving $z = -0.150/0.091 = -1.65$ ($p \approx 0.10$), which would *not* be significant at $\alpha = 0.05$.

2.3 Post-Hoc Threshold Selection

The authors state: "Various distance thresholds were tested for goodness of fit and it was found that the distance effect waned sharply after approximately 150 miles." This is a researcher degree-of-freedom problem. Testing thresholds at, say, 50, 100, 150, 200, 250, 300 miles and selecting the best-fitting one inflates the Type I error rate. A continuous distance measure or a pre-registered threshold would be more rigorous.

2.4 Untested Mechanism

The circadian rhythm explanation for the east/west asymmetry is plausible but untested. The authors argue that daytime NCAA tournament games favor Eastern teams (whose body-clock peak aligns with afternoon tip-offs), while nighttime NFL/NBA games favor Western teams. However, game time-of-day is not included as a variable, so this mechanism cannot be isolated from other explanations.

3 Our Replication

3.1 Data and Model

We use 16,689 D1 games from the 2024–2026 seasons (regular season and postseason), with the following travel features:

$$\text{travel\_advantage} = d(\text{team}_a, \text{venue}) - d(\text{team}_b, \text{venue}) \tag{2}$$

$$\text{tz\_advantage} = |\text{UTC}_a - \text{UTC}_{\text{venue}}| - |\text{UTC}_b - \text{UTC}_{\text{venue}}| \tag{3}$$

where $d(\cdot, \cdot)$ is the haversine (great-circle) distance:

$$d(\phi_1, \lambda_1, \phi_2, \lambda_2) = 2R \arcsin\left(\sqrt{\sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)}\right) \tag{4}$$

with $R = 3{,}958.8$ miles. These features enter an XGBoost model alongside 44 other features capturing team quality (prior-season efficiency, roster BPM, running margins, strength of schedule, etc.).

3.2 Result 1: Travel Distance Effect

Subset	r_{pb}	p	N
Non-neutral	-0.052	< 0.001	14,721
Neutral site	-0.024	0.297	1,968

Table 3: Point-biserial correlation between travel advantage and home win.

Travel distance is statistically significant for non-neutral games ($r^2 = 0.0027$, explaining 0.27% of variance) but **not significant for neutral-site games**—the exact context of Clay's study. The XGBoost model ranks `travel_advantage` as feature #8 of 46 by mean |SHAP| (2.7% of total), confirming it is a useful but secondary predictor.

3.3 Result 2: East/West Timezone Asymmetry

We do not replicate Clay's finding that traveling east is disproportionately harmful. Our data for non-neutral games:

Overall cross-timezone: east 33.3% vs. west 31.5%. The difference is small ($\Delta = 1.8$pp) and in the *opposite* direction from Clay. At 2+ zones, traveling west yields 28.3% vs. 32.4% for east.

TZ Crossed	Direction	Away Win %	N
0	same	36.5%	10,279
1	east	33.8%	1,738
1	west	33.4%	3,631
2	east	32.4%	207
2	west	28.3%	551
3	east	25.8%	93
3	west	25.5%	247

Table 4: Away team win rate by timezone crossing. At 2+ zones, traveling *west* has a lower win rate than east, opposite to Clay's finding.

3.4 Result 3: Quality Confounding Attenuates TZ Effects

We fit two logistic regressions predicting away wins in non-neutral games ($N = 14{,}721$):

	Model A (TZ only)		Model B (TZ + quality)	
Variable	$\hat{\beta}$	OR	$\hat{\beta}$	OR
TZ east	+0.238	1.269	+0.188	1.207
TZ west	+0.182	1.199	+0.155	1.167
Quality diff	—	—	−3.131	0.044

Table 5: Logistic regression coefficients. Adding continuous team quality reduces TZ coefficients by ∼20%.

Note the coefficient signs: positive $\hat{\beta}$ for TZ east/west means crossing *more* timezones is associated with *higher* away win rates. This is because cross-timezone games in regular-season play tend to involve quality non-conference opponents. Once quality is controlled, both coefficients shrink, and the east/west gap narrows to $\Delta\text{OR} = 0.04$ (1.207 vs. 1.167).

3.5 Result 4: Feature Importance in the Full Model

Rank	Feature	Mean |SHAP|	% Total
1	diff_run_avg_margin	0.568	19.1%
2	diff_roster_bpm	0.261	8.8%
3	diff_run_recent_avg_margin	0.179	6.0%
4	diff_sos	0.141	4.7%
⋮	⋮	⋮	⋮
8	**travel_advantage**	**0.080**	**2.7%**
⋮	⋮	⋮	⋮
36	**tz_advantage**	**0.030**	**1.0%**

Table 6: XGBoost SHAP feature importance. Travel is meaningful but secondary; timezone is near the bottom.

4 Summary of Agreements and Disagreements

Claim	Clay (2015)	Our Data
Travel distance predicts outcomes	✓(OR = 0.644)	✓($r = -0.052$, rank #8/46)
Traveling east hurts more than west	✓($p = 0.019$)	× (opposite pattern at 2+ TZ)
Elevation/temperature irrelevant	✓	✓(not in model)
150-mile threshold matters	✓	Partial (continuous effect, no threshold)
Pod era has no independent effect	✓	Not tested (different era)
Quality adequately controlled	Seed only	Seed insufficient; barthag reduces TZ effects ∼20%

Table 7: Summary comparison of Clay et al. findings versus our replication.

5 Implications for Model Development

Our analysis reveals a gap: `tz_advantage` is **identically zero for all 1,968 neutral-site games** in training data, because the feature engineering pipeline only computes timezone shift for non-neutral games (where the venue is the home team's location). This means the model cannot learn timezone effects for tournament predictions, even though `build_matchup_features()` does compute `tz_advantage` from `VENUE_COORDS` at prediction time.

Fixing this requires geocoding historical game venues and computing the timezone differential for neutral-site games in the training data. Whether this improves predictions depends on whether timezone effects are real and learnable with ∼2,000 neutral-site training examples.